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Department of Computer Science and Engineering

Date: 15/12/2025	**Test - 2**	**Max. Marks:60**
Semester: 5	**UG**	**Duration: 2 Hrs**
Course Title: DATABASE MANAGEMENT SYSTEMS		**Course Code: CD252IA**

S No	Part A (Questions)	M	BT	CO
1.	Write one major difference between JOIN and CARTESIAN PRODUCT.	1	2	3
2.	Derive INTERSECTION operator in terms of union and set difference?	1	2	3
3.	Consider the relation Student (Name, Dept, City). Write an SQL query to find all students whose names start with "S" and end with "a".	1	3	4
4.	In the COMPANY database, the relation EMPLOYEE (Fname, Minit, Lname, Ssn, Super_ssn, Dno) stores employee details. Write an SQL query to retrieve the names of all employees who do not have supervisors.	1	4	3
5.	Given relation **Employee (EID, Name, Dept, Salary)**, Write a relational algebra query to find the names of employees in the CSE department earning more than ₹50,000.	2	3	2
6.	Define the basic constraint types that are supported in SQL.	2	2	3
7.	Find out all the possible FDs in the following example:	2	3	4

A	B	C	D
a1	b2	c2	d1
a1	b2	c3	d2
a1	b1	c3	d2
a2	b3	c2	d3

S No	Part B (Questions)	M	BT	CO
1.	Consider the following relations from a university database: 1. **Book(BookID, Title, Author, Publisher, Year)** 2. **Member(MemberID, Name, Address)** 3. **Borrow(MemberID, BookID, BorrowDate, ReturnDate)** Answer the following using **relational algebra**: 1. Retrieve the names of members who have borrowed the book titled **"Database Systems."** 2. Find the titles of books borrowed by the member named **"Tom"**. 3. Retrieve the names of members who have borrowed **more than 3 distinct books.** 4. Retrieve the names of members who have **borrowed at least one book from every author** present in the Book relation.	10	4	4
2.	(a) Identify and demonstrate various additional extensions to the JOIN (OUTER JOINS) in Relational Algebra with example. (b) Explain the purpose and correct usage of the six SQL clauses (SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY).	5 5	1 1	2 2
3.	a) Find the Equivalence of sets of a Relation $R(A,B,C,D)$ with two FD sets: F={ A→B, B→C, AB→D} G={A→B, B→C,A→C,A→D} b) Find the minimal cover on the relation R(A,B,C,D,E,F) which has the following Functional dependencies: $F = \{A \rightarrow BC, C \rightarrow D, AB \rightarrow C, D \rightarrow A, A \rightarrow B\}$	5 5	4 4	3 3
4.	Specify the following queries in SQL on the COMPANY relational database schema. a. Retrieve the names of all employees in department 5 who work more than10 hours per week on the ProductX project. b. List the names of all employees who have a dependent with the same first name as themselves. c. Find the names of all employees who are directly supervised by 'Franklin Wong'. d. Find the names of all employees who work on every project controlled by department number 5.	10	4	4
5.	From the below instance of relation R(A, B, C, D): 1. Identify all non-trivial functional dependencies that hold. 2. Determine the candidate key(s) for R. 3. State the highest normal form satisfied by R	10	2	3

A	B	C	D
1	x	p	100
1	x	q	100
2	y	q	200
2	y	q	200
3	z	r	300
3	z	r	400